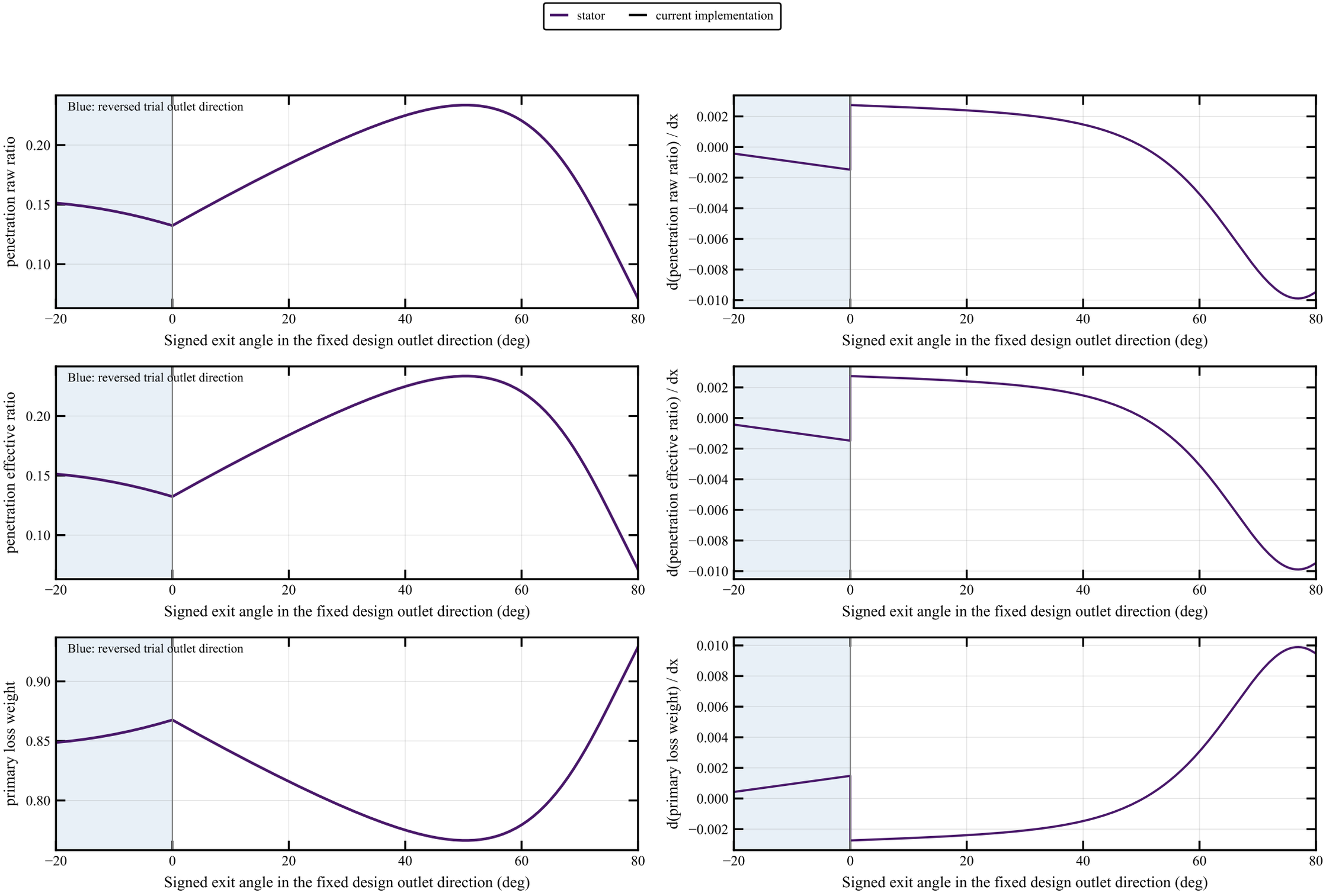benner penetration exit angle — values and first derivatives